```
 AS V1.42 Beta [Bld 211] - Source File add.asm - Page 1 - 11/23/2021 12:57:55 PM
add.asm - Copyright (C) Kevin Cole 2020.10.06 (GPL)


       1/       0 :                             PAGE    40              ; 40 lines per page
       2/       0 :                             TITLE   "add.asm - Copyright (C) Kevin Cole 2020.10.06 (GPL)"
       3/       0 :                     ;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
       4/       0 :                     ;
       5/       0 :                     ; NAME:    add.asm
       6/       0 :                     ; AUTHOR:  Kevin Cole ("The Ubuntourist") <kevin.cole@novawebdevelopment.org>
       7/       0 :                     ; LASTMOD: 2020.10.06 (kjc)
       8/       0 :                     ;
       9/       0 :                     ; DESCRIPTION:
      10/       0 :                     ;
      11/       0 :                     ;     This is the assembly source code that produces the same machine
      12/       0 :                     ;     op codes as shown by the first example (add two numbers) in the
      13/       0 :                     ;     Altair 8800 Operator’s Manual.
      14/       0 :                     ;
      15/       0 :                     ;     On a modern (as of 2020) Linux machine, install the
      16/       0 :                     ;     Macroassembler AS found at:
      17/       0 :                     ;
      18/       0 :                     ;         http://john.ccac.rwth-aachen.de:8000/as/
      19/       0 :                     ;
      20/       0 :                     ;     Then assemble this source file with the command:
      21/       0 :                     ;
      22/       0 :                     ;         asl -a -cpu 8080 -L -listradix 8 add.asm
      23/       0 :                     ;
      24/       0 :                     ;     to compare the generated list file (add.lst) to the octal machine
      25/       0 :                     ;     op codes provided in the manual.
      26/       0 :                     ;
      27/       0 :                     ;     However, if you have installed "most" -- you know you want it --
      28/       0 :                     ;     it is VERY instructive to instead assemble the source file with
      29/       0 :                     ;     the command:
      30/       0 :                     ;
      31/       0 :                     ;         asl -a -cpu 8080 -L -listradix 16 add.asm
      32/       0 :                     ;
      33/       0 :                     ;     This produces the same binary file (add.p) but changes the format
      34/       0 :                     ;     of the list file, displaying the machine op codes in hexadecimal,
      35/       0 :                     ;     rather than octal, and, together with the command:
      36/       0 :                     ;
      37/       0 :                     ;         most add.p
      38/       0 :                     ;
      39/       0 :                     ;      you can study the actual binary "executable" produced, and
      40/       0 :                     ;      "debug" it by finding the machine op codes (in hex) embedded
 AS V1.42 Beta [Bld 211] - Source File add.asm - Page 2 - 11/23/2021 12:57:55 PM
add.asm - Copyright (C) Kevin Cole 2020.10.06 (GPL)


      41/       0 :                     ;      in the executable (by comparing it to add.lst).
      42/       0 :                     ;
      43/       0 :                     ;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;;
      44/       0 :
      45/       0 :                     ; Code segment:
      46/       0 :
      47/       0 :                             ORG     0o      ; Set Program Counter to address 0
      48/       0 : 3A 80 00            START:  LDA     VAL1    ; Load value (5) at VAL1 (200) into Accumulator
      49/       3 : 47                          MOV     B,A     ; Move value in Accumulator to Register B
      50/       4 : 3A 81 00                    LDA     VAL2    ; Load value (8) at VAL2 (201) into Accumulator
      51/       7 : 80                          ADD     B       ; Add value in Register B to value in Accumulator
      52/       8 : 32 82 00                    STA     SUM     ; Store accumulator at SUM (202)
      53/       B : C3 00 00                    JMP     START   ; Jump to start of code (infinite loop)
      54/       E :
      55/       E :                     ; Data segment:
      56/       E :
      57/      80 :                             ORG     200o    ; Set Program Counter to address 200
      58/      80 : 05                  VAL1:   DB      5o      ; Data Byte at address 200 = 5
      59/      81 : 08                  VAL2:   DB      10o     ; Data Byte at address 201 = 8 (10 octal)
      60/      82 : 00                  SUM:    DB      0o      ; Data Byte at address 202 = 0
      61/      83 :
      62/      83 :                             END             ; End
 AS V1.42 Beta [Bld 211] - Source File add.asm() - Page 3 - 11/23/2021 12:57:55 PM
add.asm - Copyright (C) Kevin Cole 2020.10.06 (GPL)


  Symbol Table (* = unused):
  --------------------------

*ARCHITECTURE :                                        "x86_64-unknown-linux" - |
*BIGENDIAN :                    0 - | *BRANCHEXT :                          0 - |
*CASESENSITIVE :                0 - | *COMPMODE :                           0 - |
*CONSTPI :      3.141592653589793 - | *CUSTOM :                             0 - |
*DATE :              "11/23/2021" - | *FALSE :                              0 - |
*FULLPMMU :                     1 - | *HAS64 :                              1 - |
*HASDSP :                       0 - | *HASFPU :                             0 - |
*HASPMMU :                      0 - | *INEXTMODE :                          0 - |
*INLWORDMODE :                  0 - | *INMAXMODE :                          0 - |
*INSRCMODE :                    0 - | *INSUPMODE :                          0 - |
*LISTON :                       1 - | *MACEXP :                             7 - |
*MOMCPU :                    8080 - | *MOMCPUNAME :                    "8080" - |
*NESTMAX :                    100 - | *PACKING :                            0 - |
*PADDING :                      1 - | *RELAXED :                            0 - |
 START :                        0 C |  SUM :                               82 C |
*TIME :             "12:57:55 PM" - | *TRUE :                               1 - |
 VAL1 :                        80 C |  VAL2 :                              81 C |
*VERSION :                   142F - | *Z80SYNTAX :                          0 - |

     35 symbols
     31 unused symbols
 AS V1.42 Beta [Bld 211] - Source File add.asm() - Page 4 - 11/23/2021 12:57:55 PM
add.asm - Copyright (C) Kevin Cole 2020.10.06 (GPL)


  Code Pages:
  ----------

STANDARD (0 changed characters)

1 code page

0.00 seconds assembly time

     63 lines source file
      2 passes
      0 errors
      0 warnings
```